

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Unit Test 2

Unit Test 2 - Answer Key

Class: Class 9

Assessment Type: Unit Test 2

Total Marks: 10

Duration: 30 minutes

Topics Covered: Data Structures

ICT Teaching Resources Portal

Answer Key - Grading Guidelines

Note to Teacher: This answer key provides expected answers and marking guidelines. Accept equivalent answers where appropriate. Partial marks may be awarded for incomplete but correct responses.

Section I: MCQ (5 x 1 = 5 marks)

- Answer:** b) [1, 2, 5]
 - Explanation:** `d["a"]` references the list [1, 2]. `append(5)` adds 5 to that list, making it [1, 2, 5].
- Answer:** b) `sort()`
 - Explanation:** `sort()` sorts the list in place and modifies the original list. `sorted()` returns a new sorted list without modifying the original.
- Answer:** b) ([1, 2, 5], [3, 4])
 - Explanation:** Tuples are immutable, but the list inside the tuple is mutable. `t[0].append(5)` modifies the list at index 0, which is allowed.
- Answer:** d) Both B and C
 - Explanation:** `list[:]` and `list.copy()` both create shallow copies of the list. Option A (`copy = list`) only creates a reference, not a copy.
- Answer:** a) An arbitrary element from the set
 - Explanation:** `set.pop()` removes and returns an arbitrary element from the set. Sets are unordered, so the element returned is not predictable.

Section II: Short Answer (5 x 1 = 5 marks)

- Write the output of the sorted list slice.**
 - Answer:** Output: [9, 6, 5]
 - Explanation:** `sorted(nums, reverse=True)` gives [9, 6, 5, 4, 3, 2, 1]. `[:3]` takes the first three elements.
- Write a program to merge two dictionaries (summing values for common keys).**
 - Answer:**

```
def merge_dicts(d1, d2):
    result = d1.copy()
    for key, value in d2.items():
        if key in result:
            result[key] += value
        else:
            result[key] = value
    return result

d1 = {"a": 1, "b": 2, "c": 3}
d2 = {"b": 4, "c": 5, "d": 6}
print(merge_dicts(d1, d2))
# Output: {'a': 1, 'b': 6, 'c': 8, 'd': 6}
```

- What will be the output and why?**
 - Answer:** Output: [1, 2, 3, 4]

- **Explanation:** `lst2 = lst` makes both variables reference the same list object. When `lst2.append(4)` is called, it modifies the same list that `lst` points to.

4. **Differentiate between `pop()` and `remove()` for dictionaries.**

- **Answer:**
 - `dict.pop(key)`: Removes the key-value pair and returns the value. Can provide a default value if key not found.
 - `dict.pop(key, default)`: Same as above but returns default if key doesn't exist (no error).
 - `del dict[key]`: Removes the key-value pair. Raises `KeyError` if key not found.
- **Marking:** 1 mark for any two correct explanations, 1 mark for examples.

5. **Write a program to flatten a nested list.**

- **Answer:**

```
def flatten(nested_list):
    flat = []
    for sublist in nested_list:
        for item in sublist:
            flat.append(item)
    return flat

nested = [[1, 2], [3, 4], [5, 6, 7]]
print(flatten(nested))
# Output: [1, 2, 3, 4, 5, 6, 7]
```